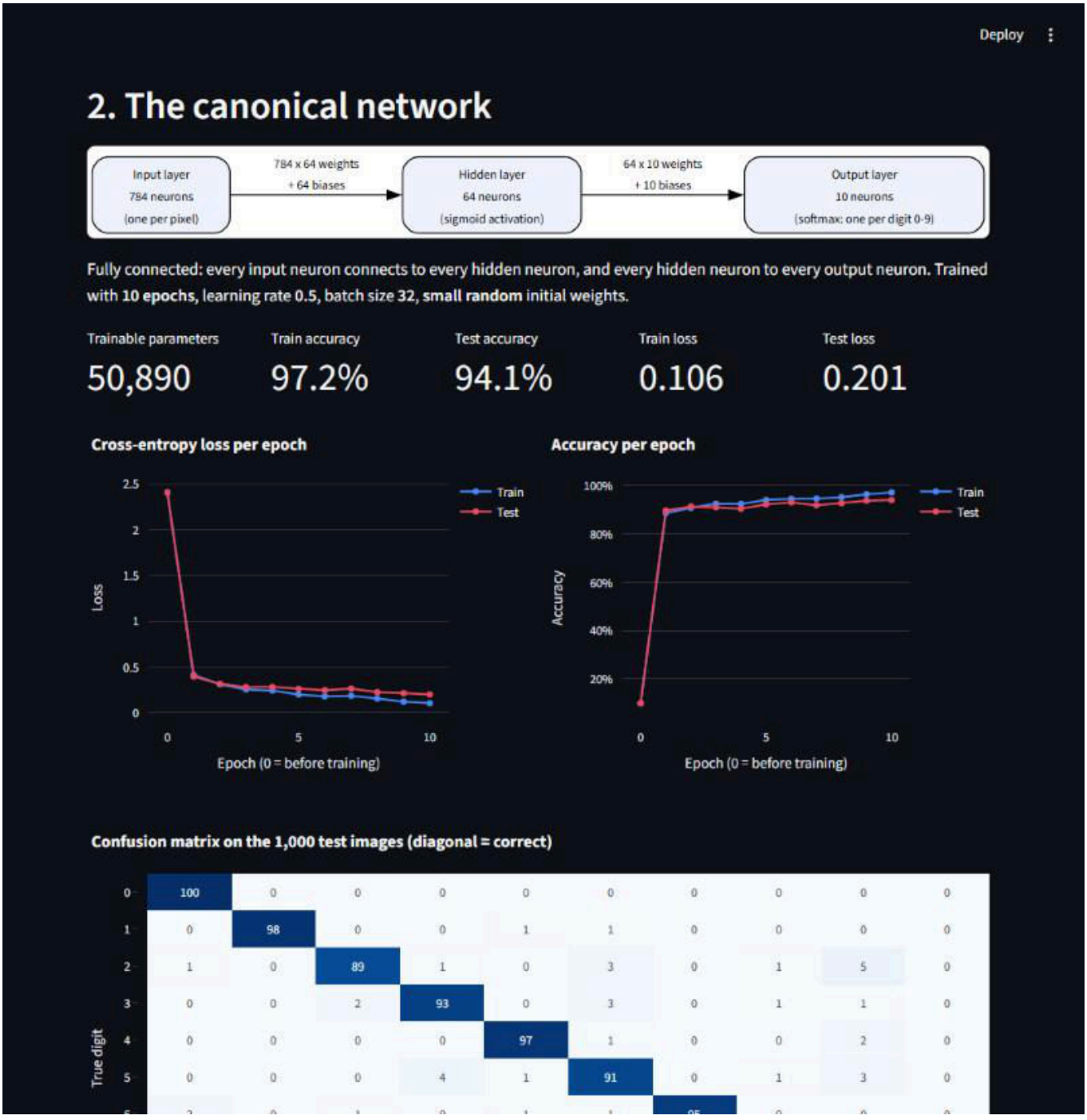
Activity 15 — Digit Lab: Neural Networks

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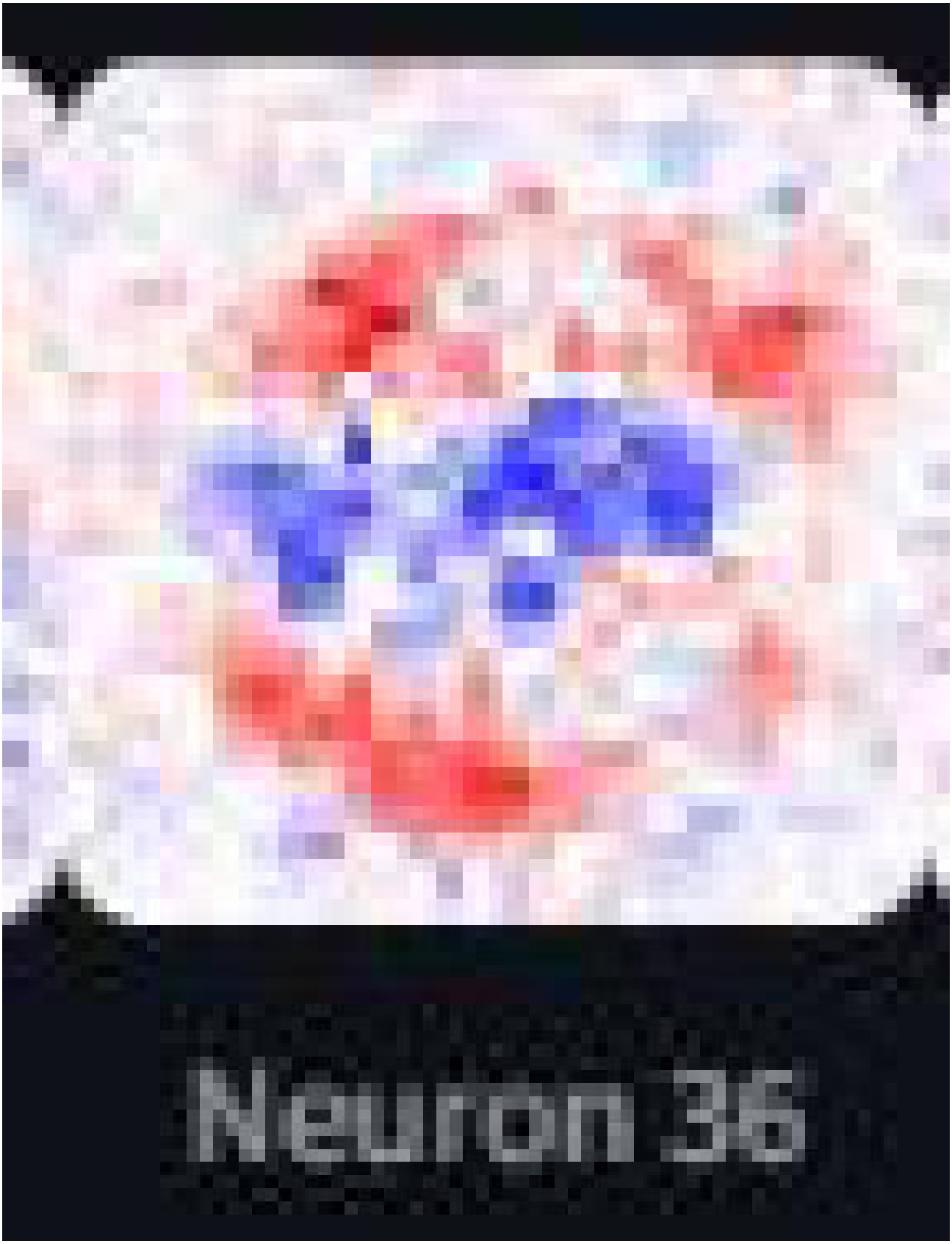
A — Canonical network (Task 1)

784 input neurons, 64 hidden, 10 output. 50,890 trainable parameters, 97.2% train and 94.1% test accuracy, losses 0.106 and 0.201.



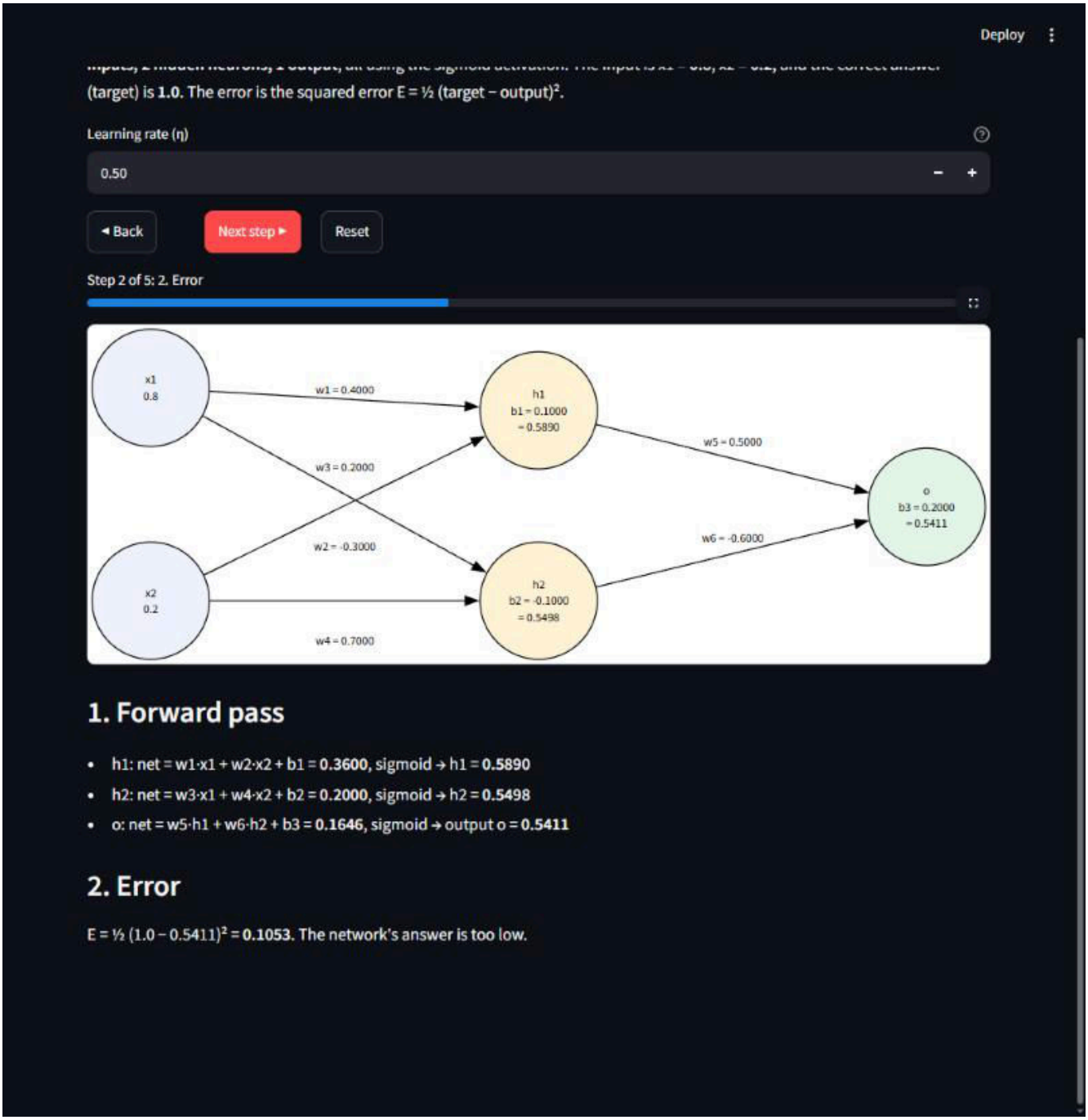
B — Hidden neuron 36 (Task 2)

Red arcs on top and bottom, blue band in the middle: it fires on a round loop with an empty center, like a 0.



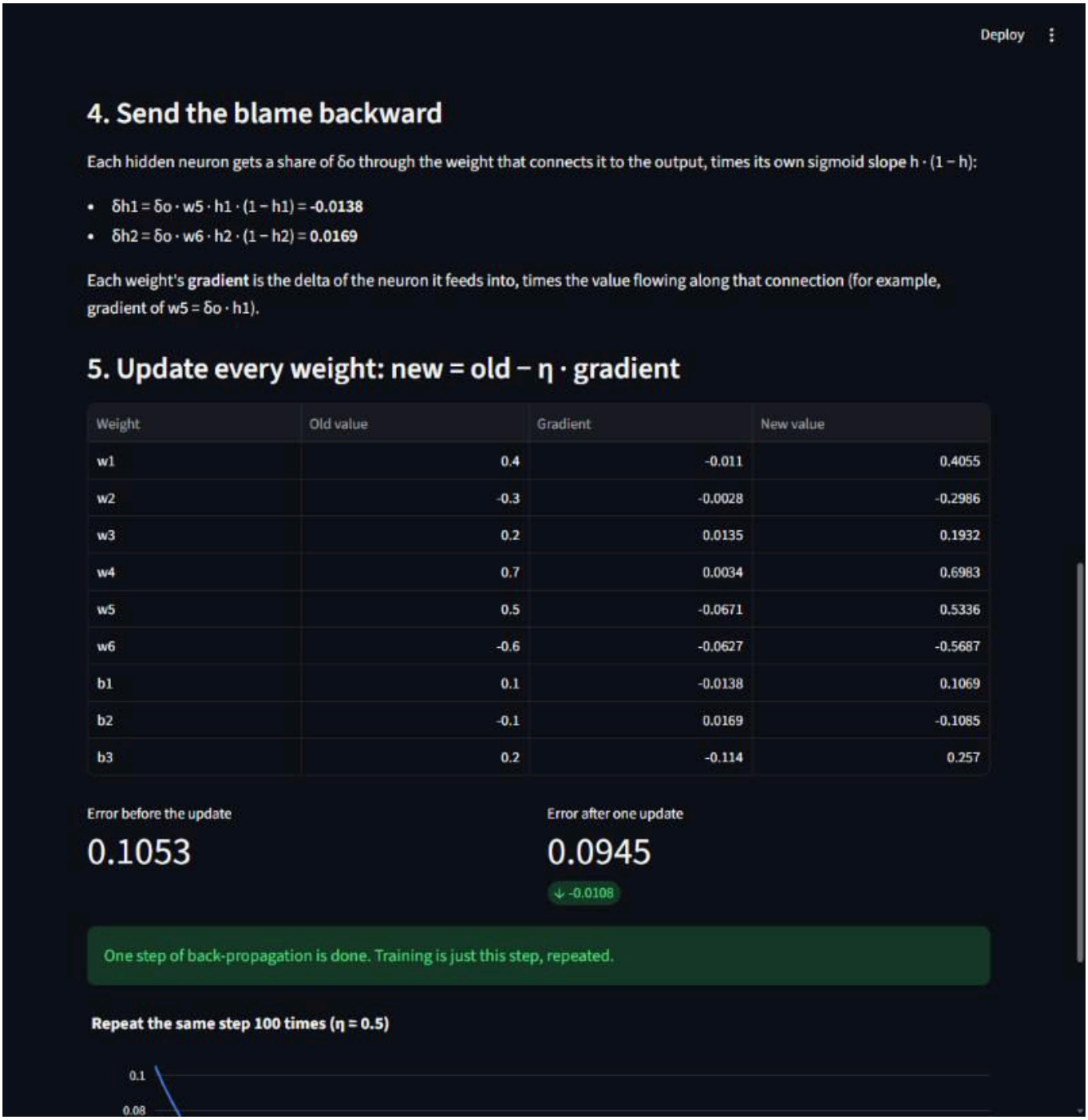
C — Back-propagation, step 2: error (Task 3)

At learning rate 0.50 the output is o = 0.5411 and the error is E = 0.1053, the answer is too low for a target of 1.0.



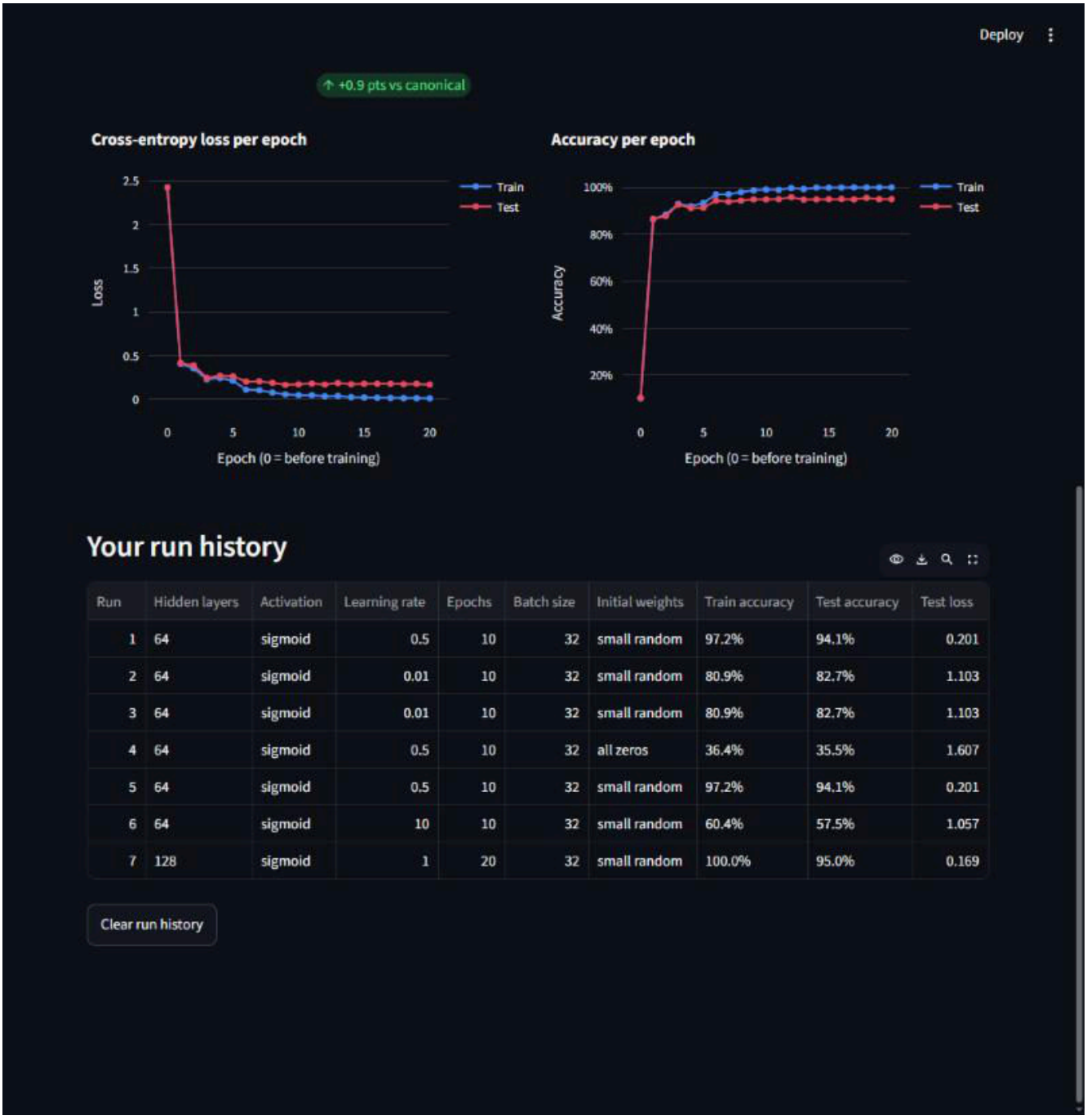
D — Back-propagation, step 5: update (Task 3)

Deltas δh1 = –0.0138 and δh2 = 0.0169; w5 goes from 0.5 to 0.5336 and the error drops to 0.0945 after one update.



E — Run history (Task 4)

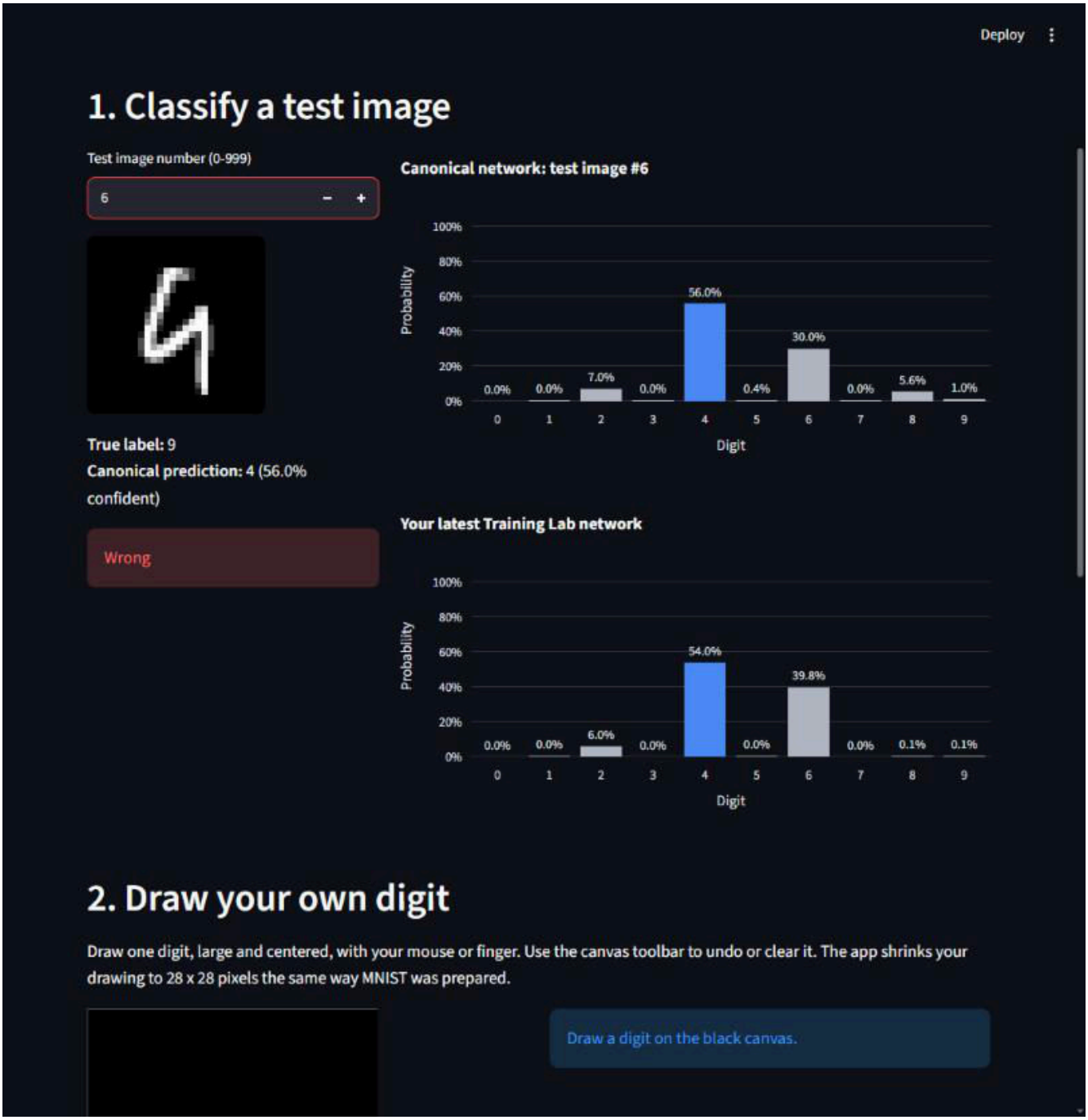
Defaults 94.1% (same as canonical), η = 0.01 82.7%, all zeros 35.5%, η = 10 57.5%, and my best run: 128 neurons, η = 1.0, 20 epochs, 95.0%.



Run	Hidden layers	Activation	Learning rate	Epochs	Batch size	Initial weights	Train accuracy	Test accuracy	Test loss
1	64	sigmoid	0.5	10	32	small random	97.2%	94.1%	0.201
2	64	sigmoid	0.01	10	32	small random	80.9%	82.7%	1.103
3	64	sigmoid	0.01	10	32	small random	80.9%	82.7%	1.103
4	64	sigmoid	0.5	10	32	all zeros	36.4%	35.5%	1.607
5	64	sigmoid	0.5	10	32	small random	97.2%	94.1%	0.201
6	64	sigmoid	10	10	32	small random	60.4%	57.5%	1.057
7	128	sigmoid	1	20	32	small random	100.0%	95.0%	0.169

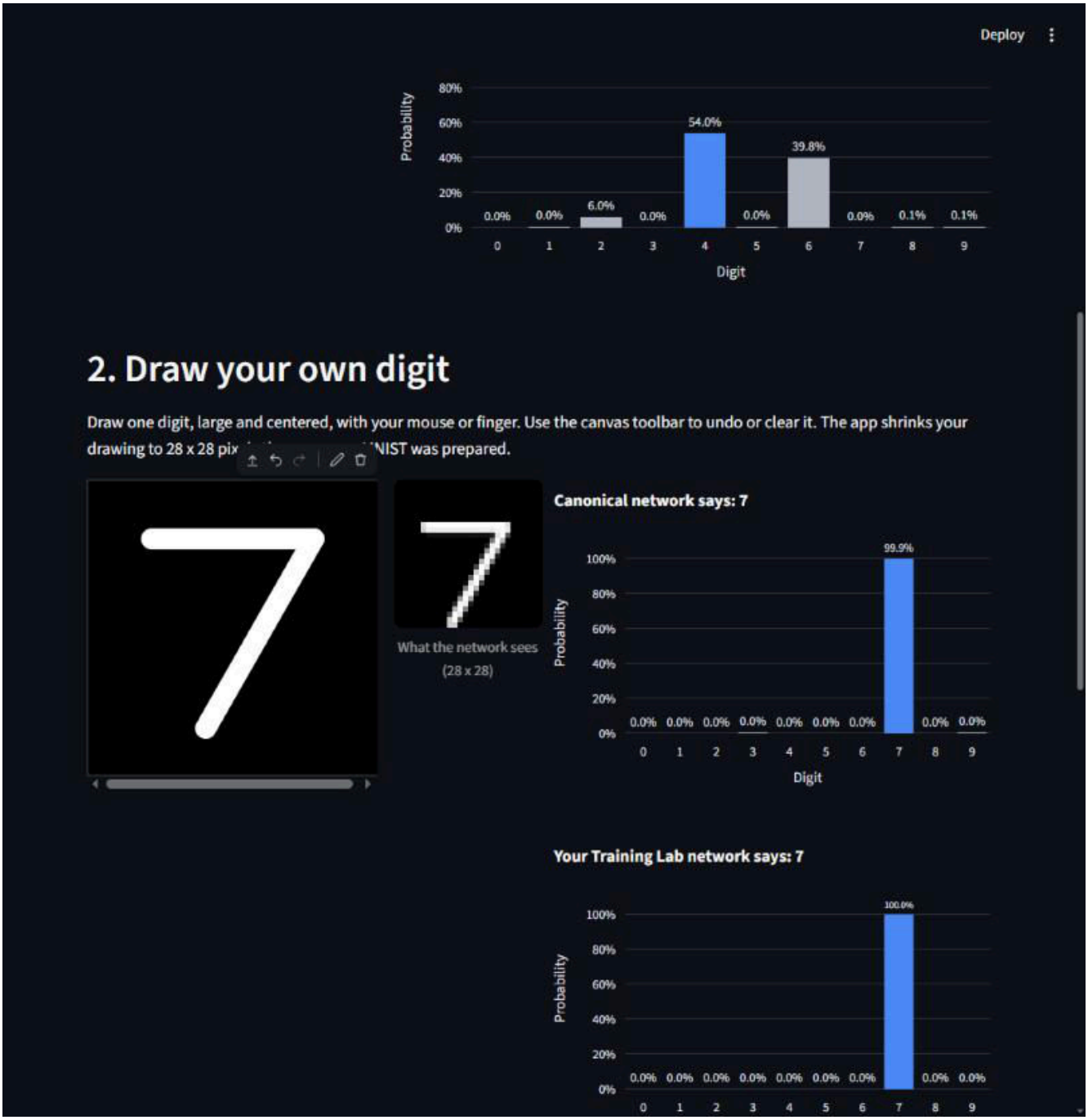
F — Test image #6 (Task 5)

True label 9, the canonical network says 4 with 56.0% (second choice 6, 30.0%). It is one of the 59 misclassified test images.



G — My drawing: 7 (Task 6)

A large centered 7, the network says 7 with 99.9%. Next to the canvas is the 28 × 28 version it actually sees.



H — My drawing: small off-center 4 (Task 6)

A small blocky 4 in the corner. The app re-centers it and the network still says 4 (93.3%), but 6.4% leaks to 9.

